Homework 4: Machine Translation

CS 1470/2470

Due November 5, 2019 at 11:59pm

1 Conceptual Questions

1. What is the purpose of the positional encoding in the Transformer architecture? (2-3 sentences)

   Answer: The positional encoding is a vector that encoder information about the position of a word in a sequence. The purpose of adding the positional encoding to the embedding to get our embedding with time signal vector.

2. What are the limitations of RNNS that Transformers solve? (3-6 sentences)

   Answer: There are several key RNN limitations. First, RNN training is hard to parallelize since the previous word must be processed before next word. Second, even with LSTMs/GRUs, preserving important linguistic context over very long sequences is difficult. The Transformer model does not have such issue: it is trivially parallelizable, and the transformers don't need to remember things(like recurrent model).

3. Consider the parameters for two different attention heads. Is it necessary that they be initialized randomly, or could we just start them all with the same vector of values? (2-4 sentences)

   Answer: I think they should be initialized randomly. Since each word has various roles and meaning in every sentences, so if all of the attention heads start with a same value, it would cause difficulty on training such model.

4. In an encoder, that has attention from right to left, and left to right) is the attention word 5 pays to word 10 the same as that word 10 pays to word 5? (3-5 sentences)

   Answer: Since the covariance(attention) matrix(attention weight) does not guarantee symmetry, so the matual attention between two words may

not be the same. And in the training process, there is no such constraints on the symmetry of such attention matrix. A word could be totally irrelavant to another word, whereas the other word could be very meaningful to the word.

5. (Optional) Have feedback for this assignment? Found something confusing? We'd love to hear from you!

   I would say the time usage in the TA hours is not very efficient: a lot of students who typically took over 30 mins to ask question, ask about the code which tremendously affect the time schedule of the following student. As for me, I ranked around 20th on Sunday's TA hours, and it took me two and half hour to wait.

2 Ethical Implications

In lecture, we discussed the environmental impacts of deep learning, and saw how transformers are some of the most energy-intensive models to train. We also saw a few suggestions for how the field could keep those factors in mind moving forward.

1. Network pruning is an active area of research within the field of deep learning. Why does the strategy described in lecture "work"? How does it have the potential to reduce the long-term environmental effects of deep learning? (2-3 sentences)

   Answer: Since 80% to 90% percent of the weights are useless to the result(weight is approximately zero), deleting the useless neurons and retraining the pruned network could be much more efficient than training the original network structure. In long term, training what is necessary is a smart way to reduce energy cost with satisfactory network performance. The energy cost for training would be down to below than 20% if the unnecessary part (80%) is pruned.

2. One suggestion for dealing with this is that researchers should report training time and hyperparameter sensitivity and consider computational cost as a factor in evaluating or pursuing new work.

   (a) What, if any, are the disadvantages of the research community moving in this direction?

   Answer: The first disadvantage would be a greater amount of energy cost, since calculating weight sensitivity would be extremely costly, especially for these models with over hundreds of thousands of hyperparameters. And maybe the value of the new model would not

compensate the value of electrics cost on training such model.

(b) What is your evaluation of how likely or feasible this suggestion is? Do you think individuals are likely to start considering these considerations on their own? Why/ why not?

Answer: For models with less than a thousands hyperparameters, this suggestion might work. In fact, uncertainty quantification is doing exactly the same thing as you suggested. The hyperparameters sensitivity is crucial in evaluating a new model for the sake of quantifying the uncertainty, not for saving energy cost. If the community is for engineering application, the cost for training should definitely be considered. I am not too optimistic if it is only for research.

3. How, if at all, does learning/thinking about the environmental consequences of the field affect how you approach deep learning? If so, how? If not, why not? (2-4 sentences) Again, no right or wrong answers here – we are looking for evidence of critical thinking, and the application of your personal beliefs and values to this problem.

Answer: My priority in research will always be creating new models to out-perform previous models, and this environmental thinking would not affect my approach in exploring deep learning. I believe the good model will always be computationally efficient and accurate, which means less energy cost and higher accuracy.